INCH-POUND

MIL-DTL-24640/5D
22 August 2002
SUPERSEDING
MIL-C-24640/5C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-PAIR, AWG 20, TYPE TTXS (INCLUDING VARIATION TYPE TTXSO)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Non-Watertight)

First - 19-strand, tin-coated copper conductor, AWG size 20.

Second - Component wires shall be MIL-W-81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be done in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling.

Third - Two component wires, one black and one white, shall be twisted together to form the required number of pairs. The pitch shall be 1 ± ¼ inch.

Fourth - Binder tape applied helically with overlap, at manufacturer's option.

Fifth - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.

Sixth - Shield insulation of two polyester tapes or polyester tape and extruded jacket. Identification codes applied using Method 2.

Seventh - The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with shielded pair number 1 from center outward. Fillers may be used as necessary, to form a round cable.

Eighth - Binder tape applied helically with overlap.

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CONSTRUCTION VARIANT FOR TYPE TTXS

Ninth - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE TTXSO

Ninth - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.
Tenth - Binder tape applied helically with overlap, at manufacturer's option.
Eleventh - Crosslinked polyolefin jacket. (See Table II for thickness).

TABLE I. Construction details.

PIN M24640/5	Type Number	Number of pairs	Jacket thickness (inch) (minimum)	Cable Diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UN	TTXS-2	2	0.038	0.335	0.361	74.3
-02UN	TTXS-4	4	.041	.393	.424	115.0

TABLE II. Construction details - overall shield.

PIN M24640/5	Type Number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UO	TTXSO-2	2	0.283	0.041	0.386	0.416	126.0
-02UO	TTXSO-6	6	.418	.044	.506	.546	219.0
-03UO	TTXSO-8	8	.510	.044	.587	.633	284.0
-04UO	TTXSO-10	10	.553	.044	.627	.675	322.0

REQUIREMENTS:

Qualification required.
Construction details: See Tables I and II.

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.

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Accelerated aging: Oven temperature, 300 ± 2°C (572 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test):
6.0 kilovolts (peak) 100 percent test.
Component wire: 8.0 kilovolts (peak), or 3 kHz spark test, 5.5 kV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 kHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum	10.95
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor and shield	1000
Conductor to shield	Not applicable
Shield to shield (component shield only)	200
Shield to shield (component shield to overall shield (TTXSO only))	500
Insulation resistance – megohms-1000 feet, minimum	
Conductor to conductor and shield	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield (TTXSO only))	100
Conductor and shield continuity	No failure
Jacket flaws (TTXSO only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	

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Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum.....50
Capacitance unbalance - percent, maximum.....15
Characteristic impedance - at 1 MHz, ohms.....30 min, 63 max
Watertightness (see Table IX of MIL-DTL-24640 for
limits of water leakage).....Not applicable
Weight .....No failure

Group B:
Cold bending .....No failure
Cross link proof test (percent maximum)
Jacket 50
Drip .....Not applicable
Tear - pounds per inch thickness, minimum 35
Physicals (unaged)
Jacket
Tensile strength - lb/in^2, minimum.....1300
Elongation - percent, minimum.....160
Attenuation - at 3 MHz, dB/100 feet, maximum 3
Gas flame (1 hour) .....Not applicable

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum.....60
Permanence of printing (conductor) - cycles, minimum .....Not applicable
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage .....No failure
Heat distortion - percent of unaged, maximum.....30
Cable sealant removability.....Not applicable
Shrinkage .....No failure
Mutual inductance - dB/1000 feet, maximum .....Not applicable
Surface transfer impedance (TTXSO only)
Milliohms per meter, maximum 700
EMP response, dB, minimum 60
Pulse response time - ms/500 feet
Rise time, maximum.....Not applicable
Decay time, maximum .....Not applicable

QUALIFICATION INSPECTION:

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Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Flame propagation	No failure
Aging and compatibility cable	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

Unit ordering length: 250 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-005)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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